

Advanced Statistical Methods II

Lecture I: Why Incomplete Data Changes Inference

Monitirtha Dey



BSDS Third Year Students
Indian Statistical Institute

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Outline of Today's Lecture

Course orientation

Motivating examples

Complete data versus observed data

Why the missingness mechanism matters

First look at imputation

Where EM will enter

Key Takeaway Messages

Where does this course fit?



Advanced Statistical Methods II is about statistical methods that go beyond the clean textbook regression/inference setup.

In many real problems, one or more of the following happens:

- ▶ the data are **incomplete, censored, or partly latent**;
- ▶ the usual squared-error or likelihood criterion is **not appropriate**;
- ▶ the estimator is defined by an objective function that is **hard to optimize directly**.

This course studies methods designed for exactly these situations.

Many modern statistical methods are built by changing one of three things.

1. Data structure

Complete observations
are replaced by
missing, latent,
censored, or
high-dimensional
observations.

2. Loss/objective

Squared error may be replaced by absolute loss, quantile loss, penalized loss, or a censored likelihood.

3. Optimization

Direct maximization/minimization may be replaced by EM, MM, Newton, or gradient-based iterations.

A large part of the course is learning how to recognize which part of the usual statistical setup has changed.

Situation → methodological response



Situation	Methodological response
Missing or latent data	EM algorithm
Hard objective function	MM algorithm / iterative optimization
Censored response	Survival likelihood, Kaplan–Meier, Cox model
Multicollinearity or many predictors	Ridge, PCR, PLS
Need sparsity	LASSO, LARS
Outliers or non-mean effects	LAD, robust regression, quantile regression
Nonlinear mean structure	Nonlinear least squares, Newton/Gauss–Newton/gradient descent

A semester in two connected arcs



Arc 1: before mid-sem

Incomplete-data thinking: missing data $\rightarrow$ imputation $\rightarrow$ EM $\rightarrow$ MM $\rightarrow$ censored survival data $\rightarrow$ Kaplan–Meier.

Arc 2: after mid-sem

Advanced regression methodology: Cox proportional hazards $\rightarrow$ regularized regression $\rightarrow$ robust/quantile regression $\rightarrow$ nonlinear regression.

	Lectures	Tutorials
Before mid-sem	12 sessions = 18 hours	6 sessions = 6 hours
After mid-sem	12 sessions = 18 hours	6 sessions = 6 hours

The tutorial in Week j will be linked to the lectures in Week j .

Suggested books and notes (till midsem)



1. Little, R. and Rubin, D. (2019). *Statistical Analysis with Missing Data*, (Third ed.). Wiley.
2. van Buuren, S. (2018). *Flexible Imputation of Missing Data* (Second ed.). Chapman and Hall/CRC. <https://stefvanbuuren.name/fimd/>.
3. Julie Josse's notes on handling missing values ([Josse, 2018](#)).

Example 1: a survey with missing income



Suppose we want to estimate average household income from a sample survey.

Person	Age	Education	Income
1	23	Graduate	42
2	45	School	?
3	36	Graduate	78
4	58	School	?
5	30	Postgraduate	95

What should we do with the two missing incomes?

Possible quick answers:

- ▶ delete rows 2 and 4;
- ▶ replace ? by the sample mean;
- ▶ predict missing income using age and education;
- ▶ write a likelihood for the observed data.

Why deletion may be misleading



Complete-case analysis keeps only observations with no missing entries.

If people with very high or very low income are less likely to report income, then complete cases are not a representative subsample.

- ▶ Missingness can change the distribution of the observed sample.
- ▶ The estimator may be biased even when the original sample was well-designed.
- ▶ The loss of observations also reduces efficiency.

The key question is not only *how much* is missing, but also *why* it is missing.

Real-world consequences



Public health: delayed COVID-19 case reporting

Public Health England reported that a technical issue meant **15,841 positive COVID-19 cases** from 25 September–2 October 2020 were not included in daily reported figures and contact-tracing data flows at the time ([Public Health England, 2020](#)).

Finance: JPMorgan's "London Whale"

The JPMorgan task-force report on the 2012 CIO losses described risk-model implementation and control failures; later summaries highlight manual Excel copy-paste workflows and a formula error affecting a Value-at-Risk calculation ([JPMorgan Chase & Co. Management Task Force, 2013](#); [Milmo, 2024](#)).

In both cases, the statistical issue is not only a missing cell. It is the chain from data capture to analysis to consequential decisions.

Example 2: hidden groups in a population



Suppose the data come from two subpopulations, but group labels are unobserved.

Observation	Measurement X	Group label Z
1	1.2	?
2	1.8	?
3	5.4	?
4	6.1	?
5	5.9	?

Statistical model

$$X_i \sim \pi N(\mu_1, \sigma_1^2) + (1 - \pi)N(\mu_2, \sigma_2^2).$$

Here the missing data are not survey nonresponses. The missing data are the **latent class labels** Z_i .

Example 3: time-to-event data



Suppose patients are followed for time to relapse.

Patient	Observed time	Relapse observed?	Meaning
1	4.2	Yes	relapse at 4.2 months
2	6.0	No	relapse time exceeds 6.0 months
3	2.7	Yes	relapse at 2.7 months
4	8.5	No	relapse time exceeds 8.5 months

A censored observation is not “missing” in the same way as a blank cell. It is partial information about an event time.

Three examples, one common structure



Problem	What is incomplete?	Later method
Survey income	Missing response values	Imputation / likelihood / EM
Mixture model	Latent group labels	EM algorithm
Relapse study	Exact failure time for censored subjects	Survival likelihood / KM

Can we write a principled likelihood using only what was observed?

Here **principled likelihood** means a likelihood for the actual random object we observed, Y_{obs} , rather than a likelihood that pretends guessed/imputed values are true.

The answer is **yes**, but the resulting observed-data likelihood is often harder to work with, because we must mathematically account for all plausible values of the unobserved part.

The complete-data likelihood is

$$L_{\text{com}}(\theta) = f(Y_{\text{obs}}, Y_{\text{mis}}; \theta), \quad \ell_{\text{com}}(\theta) = \log L_{\text{com}}(\theta).$$

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The observed-data likelihood is obtained by summing or integrating out the missing part:

$$L_{\text{obs}}(\theta) = f(Y_{\text{obs}}; \theta) = \int f(Y_{\text{obs}}, Y_{\text{mis}}; \theta) dY_{\text{mis}}.$$

$$L_{\text{obs}}(\theta) = \sum_{y_{\text{mis}}} f(Y_{\text{obs}}, y_{\text{mis}}; \theta).$$

The observed likelihood is principled, but often algebraically or computationally difficult.

The central difficulty



If complete data were known

$$\ell_{\text{com}}(\theta)$$

may be easy to maximize.

With observed data only

$$\ell_{\text{obs}}(\theta) = \log \int \exp\{\ell_{\text{com}}(\theta)\} dY_{\text{mis}}$$

may be hard to maximize.

The EM algorithm will use the simpler complete-data log-likelihood while respecting the fact that some data are unobserved.

A Normal example



Suppose

$$Y_1, \dots, Y_n \stackrel{\text{iid}}{\sim} N(\mu, \sigma^2),$$

but some Y_i values are missing.

If the complete data were observed, the MLEs are familiar:

$$\hat{\mu} = \bar{Y}, \quad \hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (Y_i - \bar{Y})^2.$$

With missing values, a tempting thought is:

“Can we somehow replace the missing sufficient statistics?”

This is the idea that will later become the E-step of EM.

Sufficient statistics viewpoint



For the Normal model, the complete-data log-likelihood depends on

$$\sum_{i=1}^n Y_i, \quad \sum_{i=1}^n Y_i^2.$$

If some observations are missing, then these sufficient statistics are also partly missing.

Preview of an EM-style idea

At iteration t , replace missing sufficient statistics by their conditional expectations given the observed data and the current parameter estimate $\theta^{(t)}$.

Do not worry about the full EM derivation today. The point is to see why the method is natural.

Introduce a missingness indicator



For each observation, define a response indicator:

$$R_i = \begin{cases} 1, & \text{if } Y_i \text{ is observed,} \\ 0, & \text{if } Y_i \text{ is missing.} \end{cases}$$

Missingness is now treated as part of the data-generating process.

The observed values alone do not tell us everything. We also need assumptions about how the missingness occurred.

The missingness mechanism



A missingness mechanism is a probabilistic model for the response pattern ([Rubin, 1976](#); [Little and Rubin, 2019](#)):

$$\mathbb{P}(R \mid Y_{\text{obs}}, Y_{\text{mis}}, \psi),$$

where ψ denotes parameters governing the missingness process.

- ▶ The data model is about Y and parameter θ .
- ▶ The missingness model is about R and parameter ψ .
- ▶ The central question is how R depends on Y_{obs} and Y_{mis} .

The classical terminology MCAR, MAR, and MNAR comes from the missing-data framework of [Rubin \(1976\)](#); see also [Little and Rubin \(2019\)](#).

Three missingness mechanisms: preview



Name	Meaning	Example intuition
MCAR	Missingness does not depend on observed or unobserved values	A lab machine randomly fails
MAR	Missingness may depend on observed values, but not on missing values after conditioning on observed values	Income missingness depends on age/education recorded in the data
MNAR	Missingness depends on the missing value itself, even after conditioning on observed values	Very high-income people are less likely to report income

These terms were formalized in the missing-data literature, especially [Rubin \(1976\)](#) and [Little and Rubin \(2019\)](#).

The distinction is about the data-generating mechanism, not just the pattern of blank cells in a spreadsheet.

Two datasets may show the same missingness pattern but have different mechanisms.

Pattern

Which entries are missing?

- ▶ monotone missingness;
- ▶ intermittent missingness;
- ▶ block missingness.

Mechanism

Why are they missing?

- ▶ random accident;
- ▶ depends on recorded covariates;
- ▶ depends on unreported values.

A statistical method is justified by assumptions about the mechanism, not by the pattern alone.

What is imputation?



Imputation means filling in missing values with plausible values so that standard complete-data methods can be applied.

Common first ideas:

- ▶ mean or median imputation;
- ▶ regression imputation;

Imputation is not the same as recovering the true missing value. It is a statistical device for inference and prediction.

Why mean imputation can be dangerous



Suppose several missing values are replaced by the observed sample mean.

What changes?

- ▶ The mean may look reasonable.
- ▶ The variance is usually underestimated.
- ▶ Correlations with other variables may be distorted.
- ▶ Standard errors are too optimistic if imputed values are treated as real observations.

A good method should reflect uncertainty about what the missing values could have been.

Conditional imputation idea



A more statistical approach uses the conditional distribution

$$Y_{\text{mis}} \mid Y_{\text{obs}}, \theta.$$

Instead of asking:

“What single value should replace the missing entry?”

we ask:

“Given what we observed, what values are plausible under the model?”

This view connects imputation, likelihood, Bayesian thinking, and EM.

The EM algorithm in one slide



The EM algorithm was introduced by [Dempster et al. \(1977\)](#) for maximum likelihood estimation from incomplete data.

Given current parameter estimate $\theta^{(t)}$:

1. **E-step:** compute the expected complete-data log-likelihood

$$Q(\theta \mid \theta^{(t)}) = \mathbb{E}_{\theta^{(t)}} [\ell_{\text{com}}(\theta) \mid Y_{\text{obs}}].$$

2. **M-step:** maximize this surrogate quantity:

$$\theta^{(t+1)} = \arg \max_{\theta} Q(\theta \mid \theta^{(t)}).$$

(Details begin in Lecture 4; today only the motivation).

Why EM feels natural



If the missing data were known	When the missing data are unknown
Write complete-data log-likelihood	Take its conditional expectation
Maximize directly	Maximize the expected complete-data log-likelihood
Obtain standard MLE update	Iterate until convergence

EM alternates between “filling in” missing information probabilistically and updating the parameter estimate.

This is not the same as single imputation; EM works at the level of expected log-likelihood or expected sufficient statistics.

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We will study MCAR, MAR, and MNAR more carefully and discuss consequences for bias, efficiency, and valid inference.

- ▶ Little and Rubin (2019, Chapter 1);
- ▶ van Buuren (2018, Chapter 2);
- ▶ Josse (2018), Section on MCAR/MAR/MNAR.

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